

Question Number	Answer	Additional Guidance	Mark
2(a)(i)	An answer that includes the following three points. • 2, to {stain / dye / colour} the chromosomes (1) • 4, to spread / separate the cells out (1) • 5, to locate and then magnify (the cells) (1)	ACCEPT to make Chromosomes visible on its own DO NOT ACCEPT stain etc cells ACCEPT so cells do not overlap ACCEPT HP to see detail	(3)

Question Number	Answer	Additional Guidance	Mark
2(a)(ii)	An explanation that includes the following points: • safety precaution identified (1) • safety precaution explained (1)	e.g. rinsing tips in water / wearing gloves / goggles because hydrochloric acid, acetic orcein is corrosive / irritant allergy to orcein	(2)

Question Number	Answer	Additional Guidance	Mark
2(b)	An answer that includes five of the following points: • use range of pH solutions (1) • (in which plants) grown (1) • (several) root tips (from each pH) (1) • take cells from same part of root tips (1) • count number of cells undergoing mitosis and total number of cells (1) • in several fields of view (1)	ACCEPT range of buffers	(5)

Question Number	Answer	Additional Guidance		Mark
2(c)(i)	A table showing the following features: - headings of pH and mitotic index (with units, %) (1) - pH data correctly entered into table (1) - mitotic index data correctly entered (1)	pH	Mitotic index (%)	(3)
		3.5	2	
		4.3	3.4	
		5.1	7.2	
		6.2	9.6	
		7.2	8.2	
		8.4	7.6	
		9	3.8	
		allow + or – 0.1		

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	An answer showing the following steps: • number of cells in mitosis identified (1) • mitotic index calculated (1) • corresponding pH value read from the graph (1)	Correct answer with no working shown gains two marks 3 e.g. $MI = (3 \div 35) \times 100 = 8.57 / 8.6 \% (1)$ 5.7 AND/OR 6.9 (ACCEPT +/- 0.05) ACCEPT correct reading from graph of any answer to mp 2	(3)

Question Number	Answer	Additional Guidance	Mark
2(c)(iii)	An explanation that includes four of the following points: • enzymes involved (1) • (enzymes are) protein (1) • 6.2 is the optimum pH (for these enzymes) (1) • at pH {below optimum / above optimum} the active site is not the right shape / enzyme denatures / bonds holding shape change (1) • enzyme substrate complexes do not form (1)		(4)